

Accessibility-in-the-Loop (AITL) Framework

Methodology Specification — Version 0.1 (Working Draft)

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1. PURPOSE

AITL is an evaluation methodology for the accessibility of AI-generated user interfaces. It applies the AI-Generated Interface Accessibility Assurance Framework (AIAAF): the evaluation framework comprising the six-dimension rubric (§4), the generative failure-pattern taxonomy, the 0–5 scoring scale with its severity-driven release gate (§5), and the element-level evidence rules. AIAAF defines what “accessible” means for AI-generated design and how to score output against that definition; AITL is the process (§6) that applies AIAAF and converts the results into guidance that design teams and tool-makers can act on before release. AITL keeps expert and assistive-technology-user judgment central; it does not treat AI as an automatic fix.

2. SCOPE AND NON-GOALS

- In scope: user interfaces produced or substantially shaped by AI systems — generated screens, components, markup, and design specifications — evaluated against WCAG 2.2 (A/AA), Section 508, and documented assistive-technology behavior.
- Out of scope (v0.1): non-UI content generation (long-form text, media), native platform apps beyond web-rendered UI, and automated remediation. AITL evaluates and guides; it does not claim to auto-fix.
- AITL complements — it does not replace — automated checkers (axe, WAVE, Lighthouse) and formal conformance audits. Its distinct contribution is the layer those tools do not cover: contextual judgment and the failure patterns specific to generative systems.

3. INPUTS

- **The artifact** in whatever form is available: rendered interface, underlying markup or component structure, design specification, or the generative prompt that produced it. More forms yield a deeper evaluation; any one form is sufficient to begin.
- **The context of use:** the user task under evaluation; assistive technologies in scope (screen reader, keyboard-only, magnification, switch access); and the conformance target (e.g., WCAG 2.2 AA).

4. THE AIAAF RUBRIC — SIX DIMENSIONS

Every evaluation scores the artifact on six dimensions: the four WCAG principles, extended with the two layers automated tools miss most. Each finding is recorded at the element level with an evidence-based rationale and a mapped success criterion.

D1 — Perceivable

Text alternatives are meaningful, not merely present; contrast meets minimums (4.5:1 normal text, 3:1 large text and UI components); structure and relationships are programmatically conveyed; media has alternatives.

D2 — Operable

Full keyboard operability without traps; visible focus with logical order; touch/click targets $\geq 24 \times 24$ CSS px; user control over motion, timing, and auto-updating content.

D3 — Understandable

Persistent, programmatically associated labels; predictable behavior; errors identified in text, programmatically associated to their field, with recovery guidance; logical reading order.

D4 — Robust

Valid, semantic markup; correct name, role, and value on all interactive elements; correct ARIA usage (no aria-hidden on focusable content); compatibility with assistive-technology parsing.

D5 — Assistive-Technology Behavior

How the output actually performs under a screen reader and keyboard against the defined task: are elements announced sensibly, is the task completable, does the experience match the visual one? This is the contextual layer conventional checkers cannot assess and is verified by hands-on AT testing, ultimately including testing with assistive-technology users.

D6 — Resistance to Generative Failure Patterns

The dimension specific to AI-generated design: a growing, versioned taxonomy of the defect classes generative systems characteristically produce. Initial taxonomy (v0.1):

- Placeholder-only labeling — fields “look” labeled but expose no accessible name.
- divAsButton — non-semantic containers with click handlers standing in for real controls.
- Generic or hallucinated alternative text — alt values produced because the attribute exists, not from the image’s purpose.
- Visual-only structure — hierarchy implied by styling but absent from markup (styled text instead of headings).
- Color-only state — errors and states conveyed by color with no text equivalent.
- Suppressed focus — outline resets for visual cleanliness with no replacement indicator.
- Low-contrast default aesthetics — light-gray secondary text below contrast minimums.
- Decorative ARIA misuse — aria-hidden or roles applied in ways that hide interactive content from AT.

The taxonomy is intended to grow through application and to be validated and published; each entry records the pattern, why generative systems produce it, its detection cue, and its remediation.

5. SCORING

Each dimension is scored 0–5 with element-level findings recorded beneath it. Findings carry a severity that drives the release gate.

Score	Label	Meaning
5	Conformant	No defects found on this dimension for the evaluated task; meets the conformance target.
4	Minor issues	Defects exist but do not impede task completion with assistive technology; low-severity fixes.
3	Degraded	Task completable with assistive technology but with significant friction or workarounds.
2	Barrier	At least one defect prevents some users from completing the task on this dimension.
1	Blocking	Multiple defects; the task is effectively unavailable to assistive-technology users.
0	Not evaluable	Artifact insufficient to assess this dimension; noted, not averaged.

Severity	Definition	Gate effect
Critical	Prevents task completion for screen-reader	Fails the release gate.

	or keyboard-only users.	
High	Major friction or exclusion for a user group; task completable with difficulty.	Fails gate unless documented exception.
Medium	Conformance failure with available workaround.	Must be scheduled for remediation.
Low	Best-practice deviation; no conformance failure.	Advisory.

6. PROCESS

- **Step 1 — Frame:** define the task, AT in scope, and conformance target.
- **Step 2 — Sweep:** run automated checks to clear the mechanically detectable layer.
- **Step 3 — Evaluate:** score D1–D6 element by element, with keyboard and screen-reader passes for D5 and the taxonomy for D6; every finding gets a rationale grounded in specific evidence and a mapped criterion.
- **Step 4 — Route:** separate what is decidable from what requires human or AT-user judgment, and route the latter explicitly.
- **Step 5 — Return signals:** produce the score profile, prioritized findings, upstream prompt/specification guidance, and the gate determination.

7. OUTPUTS

- A six-dimension score profile with a conformance determination against the stated target.
- A prioritized, element-level findings list (criterion, severity, rationale, fix).
- Upstream guidance: prompt- and specification-level requirements that steer future generation away from the defects found.
- A release-gate result (Critical findings block).
- A documented trace suitable for conformance evidence and research use.

8. VALIDATION ROADMAP

- Phase 1 (current): apply the rubric to synthetic and publicly available AI-generated interfaces; refine dimension definitions and the D6 taxonomy. A completed worked example accompanies this specification.
- Phase 2: inter-rater exercises with accessibility practitioners to test that the rubric scores consistently across evaluators.
- Phase 3: validation sessions with assistive-technology users to test that rubric findings predict real task success and failure.
- Phase 4: dissemination through peer-reviewed human-computer-interaction venues and professional channels; publication of the taxonomy for field adoption.

9. RELATIONSHIP TO EXISTING WORK

AITL builds on established standards (WCAG 2.2, Section 508, WAI-ARIA) and on the general “human-in-the-loop” evaluation tradition. Its specific contribution is narrower and deeper: an element-level evaluation rubric purpose-built for AI-generated user interfaces, and a named, growing taxonomy of generative accessibility failure patterns — the layer that neither automated checkers nor generic conformance audits currently address.

10. VERSION HISTORY

- v0.1 (July 2026) — initial specification: six-dimension rubric, 0–5 scale, severity gate, eight-pattern D6 taxonomy, five-step process, validation roadmap. Companion artifact: AITL Worked Example (sign-up screen evaluation).